Theoretical Modeling and Design Framework of Integrated Silicon Photonic Sub-Circuits for Group Index Extraction and Resonator Characterization

Nikos Kostopoulos
edX Public Username: `n-kostopoulos`
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Abstract

This work presents a comprehensive theoretical formulation and design analysis of passive silicon photonic sub-circuits implemented on a standard 220 nm silicon-on-insulator (SOI) platform (`EBeam_n-kostopoulos_v3.gds`). To establish an accurate modeling framework prior to physical fabrication and probe-station characterization, analytical models and numerical eigenmode solutions are developed across all implemented strip waveguide configurations: the standard C-band 1550 nm quasi-TE mode (500×220 nm), the C-band quasi-TM mode (500×220 nm), and the O-band 1310 nm quasi-TE mode (350×220 nm). Furthermore, circuit-level numerical simulations in Lumerical INTERCONNECT are formulated for unbalanced Mach-Zehnder interferometers (MZIs) to verify Free Spectral Range (FSR) scaling and group index (n_g) extraction. Resonator coupled-mode dynamics in add-drop and terminated double-bus microring topologies, waveguide Bragg gratings, multi-mode interference (MMI) waveguide crossing cascades, and focusing sub-wavelength grating couplers (SWGCs) are systematically evaluated.

1 Introduction and Theoretical Objectives

Silicon photonics exploits the mature infrastructure of CMOS fabrication to realize high-density photonic integrated circuits (PICs) [1]. Owing to the high refractive index contrast between the silicon core ($n_{\mathrm{Si}} \approx 3.47$) and the silicon dioxide cladding ($n_{\mathrm{SiO_2}} \approx 1.44$), optical fields can be confined within sub-micrometer dimensions. However, this tight spatial confinement induces pronounced material and structural waveguide dispersion. Optical pulses propagating through such media travel at the group velocity $v_g = c/n_g$, governed by the group index $n_g(\lambda)$. A precise theoretical framework is therefore necessary to predict modal parameters and resonance dynamics across both C-band (1550 nm) and O-band (1310 nm) regimes.

2 Silicon Strip Waveguides and Multi-Modal Dispersion

The core building block of the photonic circuitry consists of strip waveguides defined with a nominal thickness of $h = 220$ nm and nominal widths of $w = 500$ nm for 1550 nm quasi-TE and quasi-TM operation, as well as $w = 350$ nm for 1310 nm quasi-TE operation. Within these strip geometries, the monochromatic optical electric field propagating along the z-axis satisfies the planar wave relation:

$$E(x, y, z, t) = \mathcal{E}(x, y)\, e^{i(\omega t - \beta z)} \tag{1}$$

where $\mathcal{E}(x, y)$ represents the transverse spatial mode distribution, ω is the angular optical frequency, and β denotes the modal propagation constant:

$$\beta(\lambda) = \frac{2\pi}{\lambda} n_{\mathrm{eff}}(\lambda) = \frac{\omega}{c} n_{\mathrm{eff}}(\omega) \tag{2}$$

Because the effective index exhibits strong wavelength dependence due to both the intrinsic chromatic dispersion of silicon and the dielectric interface boundary conditions, the dispersion slope $dn_{\mathrm{eff}}/d\lambda$ significantly deviates from zero. Applying the definition of group velocity $v_g = d\omega/d\beta = c/n_g$, differentiation yields the group index:

$$n_g(\lambda) = n_{\mathrm{eff}}(\lambda) - \lambda \frac{dn_{\mathrm{eff}}}{d\lambda} \tag{3}$$

2.1 Modal Profile Analysis

Finite-difference eigenmode (FDE) simulations were performed in Lumerical MODE for all three distinct waveguide implementations present on the chip layout, as illustrated in Fig. 1:

1. **C-band Quasi-TE (500×220 nm):** The fundamental TE mode at $\lambda_0 = 1550$ nm exhibits strong horizontal confinement with effective index $n_{\mathrm{eff}} \approx 2.447$ and group index $n_g \approx 4.204$ [Fig. 1(a)].

2. **C-band Quasi-TM (500×220 nm):** The fundamental TM mode at $\lambda_0 = 1550$ nm features vertical field

polarization with higher evanescent leakage into the oxide cladding, yielding $n_{\text{eff}} \approx 1.769$ and $n_g \approx 3.721$ [Fig. 1(b)].

3. **O-band Quasi-TE (350×220 nm):** Reducing the core width to 350 nm shifts the cutoff wavelength of higher-order transverse modes to shorter wavelengths, preserving strictly single-mode quasi-TE operation across the O-band. At $\lambda_0 = 1310$ nm the fundamental mode exhibits $n_{\text{eff}} \approx 2.430$ and $n_g \approx 4.510$ [Fig. 1(c)].

2.2 Wavelength Dependence and Compact Models

To enable full circuit simulations in Lumerical INTERCONNECT, frequency sweeps were executed across the respective operating bands, as shown in Fig. 2. Around each design wavelength, the dispersion curves are fitted to second-order polynomial compact models:

$$n_{\text{eff}}(\lambda) \approx a_0 + a_1(\lambda - \lambda_0) + a_2(\lambda - \lambda_0)^2 \tag{4}$$

The group index at the expansion point and its slope follow directly from Eq. (4): $n_g(\lambda_0) = a_0 - \lambda_0 a_1$ and $dn_g/d\lambda = -2\lambda a_2$. Because the group index bends noticeably over the full sweep (Fig. 2), the coefficients are valid only locally, within a window of a few tens of nanometers around λ_0. The extracted numerical parameters are summarized in Table 1. The small imaginary component ($\text{Im}\{n_{\text{eff}}\} \sim 10^{-9}$) observed in numerical eigenmode plots arises solely as a solver residual artifact in lossless dielectric media.

Table 1: Modal Parameters and Local Compact Model Coefficients

Waveguide Type	λ_0 (μm)	a_0 (n_{eff})	a_1 (μm^{-1})	a_2 (μm^{-2})	n_g
Strip TE (500×220 nm)	1.55	2.447	-1.133	-0.055	4.204
Strip TM (500×220 nm)	1.55	1.769	-1.259	$+1.940$	3.721
Strip TE (350×220 nm)	1.31	2.430	-1.588	-0.050	4.510

3 Unbalanced Mach-Zehnder Interferometers (MZIs)

Interferometric architectures provide a direct, highly sensitive mechanism for mapping phase differences into measurable optical intensity modulations [1]. In this layout, unbalanced Mach-Zehnder Interferometers (`MZI1` and `MZI2`) are implemented.

3.1 Field Propagation and Superposition

An incoming optical field E_i is first split into two equal parts by an input 3 dB splitter (`ebeam_y_1550` Y-branch):

$$E_1 = E_2 = \frac{E_i}{\sqrt{2}} \tag{5}$$

The split beams then propagate through two distinct waveguide arms having physical path lengths L_1 and $L_2 = L_1 + \Delta L$. Each arm comprises concatenated routing segments and bends. The optical phase accumulated along each arm is $\phi_k = \beta_k L_k$. Assuming identical waveguide cross-sections such that $\beta_1 = \beta_2 = \beta$, and incorporating the linear power attenuation coefficient α, the transmitted electric fields immediately preceding the combiner stage are:

$$E_{o1} = \frac{E_i}{\sqrt{2}} e^{-i\beta L_1 - \frac{\alpha}{2} L_1}, \quad E_{o2} = \frac{E_i}{\sqrt{2}} e^{-i\beta L_2 - \frac{\alpha}{2} L_2} \tag{6}$$

At the output combiner (`ebeam_bdc_te1550` broadband directional coupler), the fields undergo vector summation. While the quadrature coupling phase shifts the fringe locations between the complementary through and cross ports, the fringe spectral period remains identical. For the constructive in-phase output:

$$E_o = \frac{E_{o1} + E_{o2}}{\sqrt{2}} = \frac{E_i}{2} e^{-i\beta L_1 - \frac{\alpha}{2} L_1} \left[1 + e^{-i\beta \Delta L - \frac{\alpha}{2}\Delta L}\right] \tag{7}$$

3.2 Spectral Transfer Function and n_g Extraction

Under the condition of negligible differential propagation loss ($\alpha \Delta L \ll 1$), the transmitted intensity $I_o = |E_o|^2$ normalized by the incident intensity $I_i = |E_i|^2$ is described by the sinusoidal transfer function:

$$T(\lambda) = \frac{I_o}{I_i} = \frac{1}{2}\left[1 + \cos(\beta(\lambda)\Delta L)\right] = \cos^2\left(\frac{\pi n_{\text{eff}}(\lambda)\Delta L}{\lambda}\right) \tag{8}$$

Transmission extrema arise when the accumulated optical phase difference $\Delta\phi(\lambda) = \beta(\lambda)\Delta L$ crosses integer multiples of π. Specifically, constructive interference peaks satisfy $\Delta\phi(\lambda_m) = 2m\pi$, while transmission nulls correspond to $\Delta\phi = (2m+1)\pi$.

Because $\beta(\lambda) = 2\pi n_{\text{eff}}(\lambda)/\lambda$ decreases monotonically with wavelength, the phase difference $\Delta\phi(\lambda)$ decreases with increasing λ. Consequently, between two consecutive transmission peaks located at λ_1 and λ_2 (where $\lambda_2 > \lambda_1$ and $\text{FSR} = \Delta\lambda = \lambda_2 - \lambda_1$), the phase change satisfies:

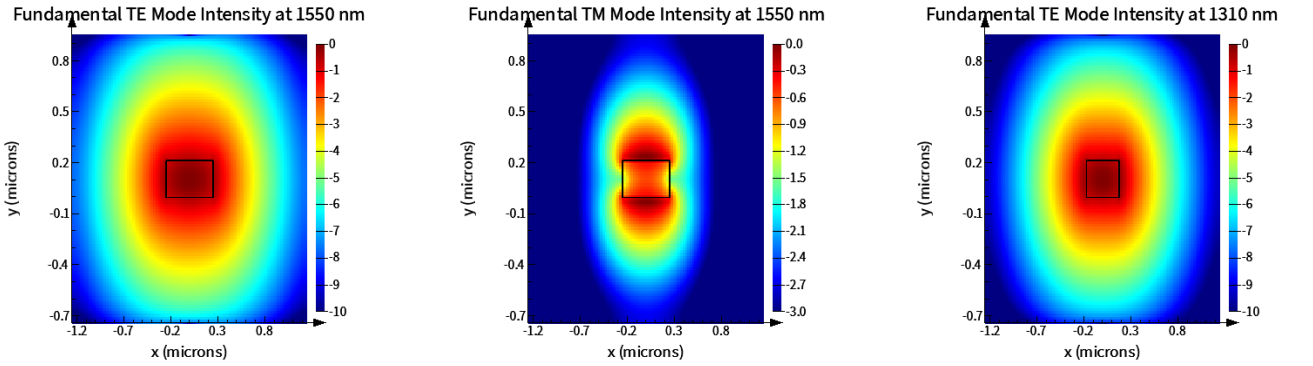
$$\Delta\phi(\lambda_1) - \Delta\phi(\lambda_2) = 2\pi \tag{9}$$

Applying a first-order Taylor expansion around the central wavelength $\lambda_0 = (\lambda_1 + \lambda_2)/2$:

$$\left|\frac{d(\Delta\phi)}{d\lambda}\right| \Delta\lambda = 2\pi \tag{10}$$

Differentiating the phase function directly with respect to wavelength:

$$\frac{d}{d\lambda}\left(\frac{2\pi n_{\text{eff}}}{\lambda}\Delta L\right) = 2\pi\Delta L\left[\frac{1}{\lambda}\frac{dn_{\text{eff}}}{d\lambda} - \frac{n_{\text{eff}}}{\lambda^2}\right] = -\frac{2\pi\Delta L}{\lambda^2} n_g(\lambda) \tag{11}$$



(a) 1550 nm Strip TE (500 × 220 nm) (b) 1550 nm Strip TM (500 × 220 nm) (c) 1310 nm Strip TE (350 × 220 nm)

Figure 1: Simulated electric field intensity profiles ($|E|^2$, log scale) calculated using the finite-difference eigenmode (FDE) solver in Lumerical MODE for all three waveguide types incorporated into the submission layout.

Taking the absolute value and substituting Eq. (11) into Eq. (10) yields the first-order relationship used for parameter extraction, valid provided n_g varies slowly over one fringe spacing:

$$n_g(\lambda_0) = \frac{\lambda_0^2}{\Delta L \cdot \mathrm{FSR}} \tag{12}$$

For our design, net path imbalances are defined by the KLayout layout geometry as $\Delta L = 174 - 34 = 140\,\mu$m for `MZI1` and $\Delta L = 204 - 34 = 170\,\mu$m for `MZI2` (accounting for top arm length differences while bottom symmetric routing segments perfectly cancel out), corresponding to nominal theoretical target FSRs of $\mathrm{FSR}_1 \approx 4.08$ nm and $\mathrm{FSR}_2 \approx 3.36$ nm at $\lambda_0 = 1550$ nm.

3.3 Circuit Simulation Results and Verification

Circuit-level optical simulations were conducted in Lumerical INTERCONNECT via SiEPIC-Tools. The Optical Network Analyzer (ONA) simulated the transmission spectra of both unbalanced interferometers across the C-band, as illustrated in Fig. 3.

In `MZI1` [Fig. 3(a)], the simulated adjacent constructive peaks near 1550 nm are located at $\lambda_1 = 1550.00$ nm and $\lambda_2 = 1554.08$ nm, corresponding to $\mathrm{FSR}_1 = 4.08$ nm. In `MZI2` [Fig. 3(b)], the increased imbalance ($\Delta L \approx 170\,\mu$m) compresses the fringe period to $\lambda_1 = 1550.00$ nm and $\lambda_2 = 1553.36$ nm, yielding $\mathrm{FSR}_2 = 3.36$ nm. Evaluating Eq. (12) with these simulated fringe spacings and consistent units yields $n_g = (1.55\,\mu\mathrm{m})^2/(140\,\mu\mathrm{m} \times 0.00408\,\mu\mathrm{m}) \approx 4.206$ and $(1.55\,\mu\mathrm{m})^2/(170\,\mu\mathrm{m} \times 0.00336\,\mu\mathrm{m}) \approx 4.206$. The slight difference of 0.002 from the eigenmode value ($n_g = 4.204$) stems directly from rounding the fringe peak separation to three significant digits.

4 Microring Resonators (Add-Drop and Terminated Double-Bus Configurations)

Optical microring resonators provide high spectral selectivity and field enhancement within a compact footprint [3]. In this work, four-port quasi-TE add-drop filters built from `ebeam_dc_halfring_straight` couplers (`Ring1`, `Ring2`) and quasi-TM rings built from the `DoubleBus_Ring` cell with terminated drop buses (`SingleBusRing1`, `SingleBusRing2`) are modeled.

4.1 Coupled-Mode Dynamics and Coupling Coefficients

Energy exchange between the straight bus waveguides and the curved ring resonator is governed by directional couplers. For an ideal, lossless coupling region, the field transmission coefficient t (through-coupling) and coupling coefficient κ (cross-coupling) obey unitary energy conservation:

$$|t|^2 + |\kappa|^2 = 1 \tag{13}$$

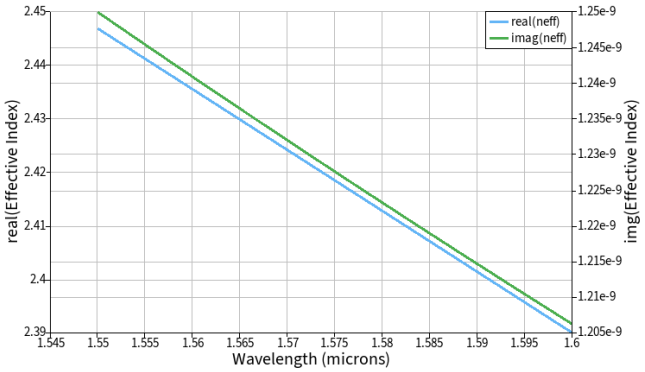
According to coupled-mode theory (CMT), κ exhibits an exponential decay with increasing gap:

$$\kappa(g) \propto \exp(-\gamma\, g) \tag{14}$$

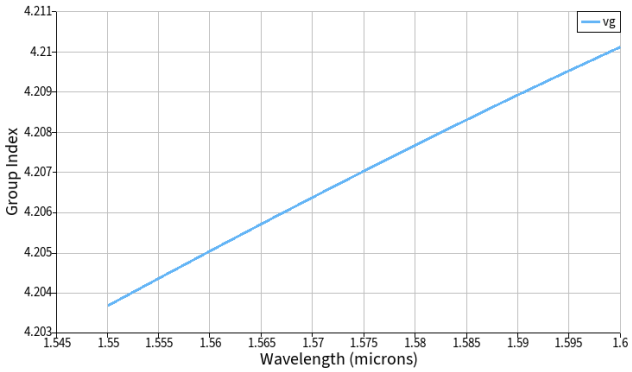
where $\gamma = \sqrt{\beta^2 - k_0^2 n_{\mathrm{clad}}^2}$ denotes the lateral evanescent decay constant. For the quasi-TM mode, its lower effective index ($n_{\mathrm{eff}} \approx 1.769$ vs. 2.447 for TE) results in a smaller lateral decay constant γ, producing a slower exponential decay of κ with coupler gap; thus, gap variations between 200 nm and 250 nm provide smooth coupling control.

4.2 Resonance Condition and Transfer Equations

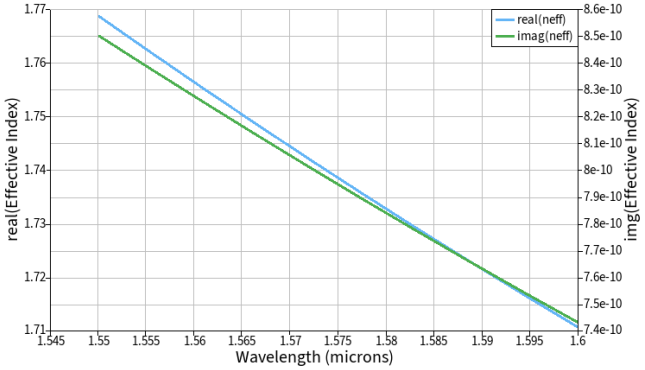
A resonance is established when the round-trip phase accumulation ϕ_{rt} around the perimeter L_{rt} matches an



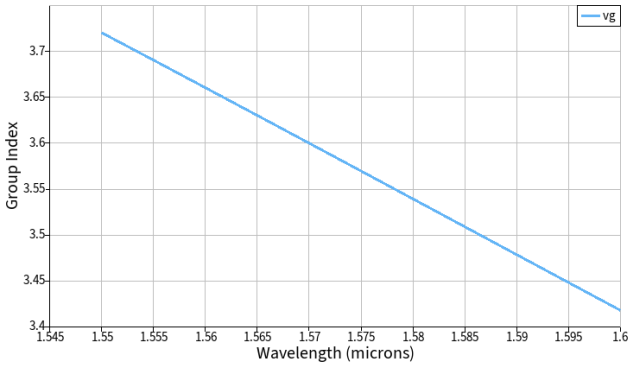
(a) n_{eff} vs. λ (1550 nm Strip TE)



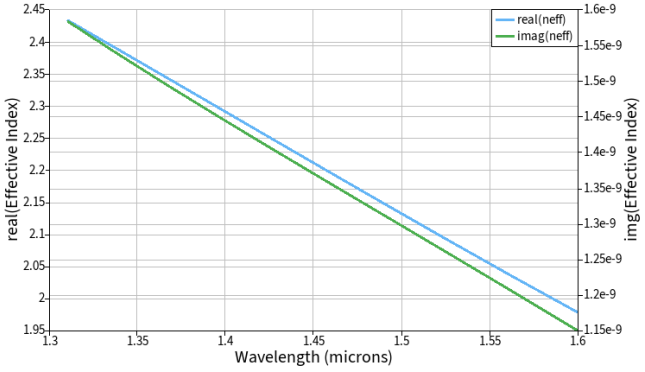
(b) n_g vs. λ (1550 nm Strip TE)



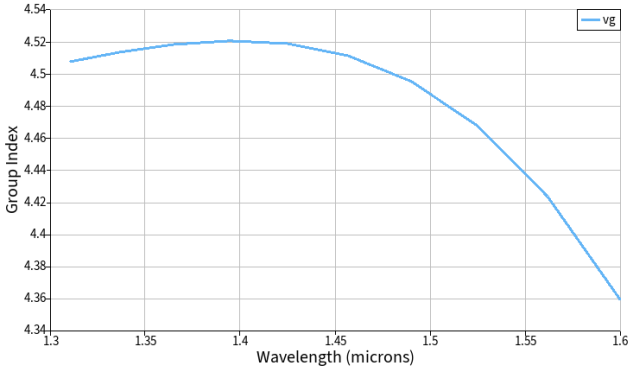
(c) n_{eff} vs. λ (1550 nm Strip TM)



(d) n_g vs. λ (1550 nm Strip TM)

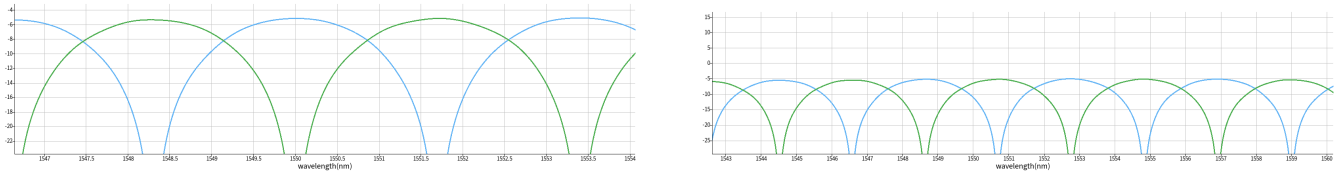


(e) n_{eff} vs. λ (1310 nm Strip TE)



(f) n_g vs. λ (1310 nm Strip TE)

Figure 2: Comprehensive dispersion characterization across all three implemented silicon strip waveguide geometries, showing effective refractive index (n_{eff}) and group index (n_g) as functions of wavelength obtained from numerical Lumerical MODE eigenmode sweeps.



(a) `MZI1` ($\Delta L \approx 140\,\mu$m, simulated FSR ≈ 4.08 nm)

(b) `MZI2` ($\Delta L \approx 170\,\mu$m, simulated FSR ≈ 3.36 nm)

Figure 3: Calculated optical transmission spectra for both unbalanced Mach-Zehnder Interferometers simulated in Lumerical INTERCONNECT. The through (blue) and cross (green) ports exhibit complementary anti-phase sinusoidal oscillations with distinct fringe spacings, verifying the FSR design models.

integer multiple of 2π:

$$\phi_{\mathrm{rt}}(\lambda) = \beta(\lambda) L_{\mathrm{rt}} = m \cdot 2\pi \implies m\lambda_m = n_{\mathrm{eff}}(\lambda_m) L_{\mathrm{rt}} \quad (15)$$

where m is the azimuthal mode order. For the add-drop cells `Ring1` and `Ring2`, the PCell parameters specify $L_c = 0$, ensuring strictly circular geometries where $L_{\mathrm{rt}} = 2\pi R$.

Adopting the forward-propagating phase factor $e^{-i\beta z}$, let $a = \exp(-\alpha L_{\mathrm{rt}}/2) = \sqrt{A}$ denote the round-trip field amplitude transmission factor (where A represents round-trip power transmission). As a reference, for an all-pass (single-bus) resonator the steady-state complex field transmission is [3]:

$$\frac{E_{\mathrm{thru}}}{E_{\mathrm{in}}} = \frac{t - a\, e^{-i\phi_{\mathrm{rt}}}}{1 - t^* a\, e^{-i\phi_{\mathrm{rt}}}} \quad (16)$$

Critical coupling occurs when the internal cavity round-trip amplitude loss matches the through-coupling transmission amplitude ($t = a$).

For an add-drop (double-bus) resonator with input and drop directional coupler coefficients (t_1, κ_1) and (t_2, κ_2), the transfer functions at the through and drop ports are expressed as [3]:

$$\frac{E_{\mathrm{thru}}}{E_{\mathrm{in}}} = \frac{t_1 - t_2^* a\, e^{-i\phi_{\mathrm{rt}}}}{1 - t_1^* t_2^* a\, e^{-i\phi_{\mathrm{rt}}}} \quad (17)$$

$$\frac{E_{\mathrm{drop}}}{E_{\mathrm{in}}} = \frac{-\kappa_1^* \kappa_2 \sqrt{a}\, e^{-i\phi_{\mathrm{rt}}/2}}{1 - t_1^* t_2^* a\, e^{-i\phi_{\mathrm{rt}}}} \quad (18)$$

On resonance, complete extinction at the through port requires $t_1 = t_2^* a$. For a perfectly symmetric filter ($t_1 = t_2$) this can only be satisfied in the lossless limit $a \to 1$; in the presence of propagation loss, symmetric couplers leave a finite through-port residual, and complete through extinction necessitates $|t_2| > |t_1|$.

In the `SingleBusRing` cells, the second bus is terminated via `ebeam_terminator_tm1550`. Consequently, no power is collected from the drop channel and the sub-circuit behaves as an add-drop filter with an unmeasured drop port rather than a true all-pass cavity. For symmetric directional couplers ($t_1 = t_2 = t$), the on-resonance through-port transmission evaluates to:

$$T_{\mathrm{min}} = \left[\frac{t\,(1-a)}{1 - t^2 a}\right]^2 \quad (19)$$

which is a monotonically increasing function of t. Stronger coupling (smaller gap, smaller t) therefore yields a deeper through-port dip and a lower loaded Q, while no zero-transmission critical-coupling point exists for $a < 1$.

The Free Spectral Range between adjacent resonant wavelengths is:

$$\mathrm{FSR} = \frac{\lambda_0^2}{n_g L_{\mathrm{rt}}} = \frac{\lambda_0^2}{2\pi R\, n_g} \quad (20)$$

For $R = 10\,\mu$m ($L_{\mathrm{rt}} \approx 62.83\,\mu$m) and $R = 5\,\mu$m ($L_{\mathrm{rt}} \approx 31.42\,\mu$m) in quasi-TE mode ($n_g \approx 4.204$), the expected FSRs evaluate to ≈ 9.10 nm and ≈ 18.19 nm, respectively. The ratio $\mathrm{FSR}(R = 5\,\mu\mathrm{m})/\mathrm{FSR}(R = 10\,\mu\mathrm{m})$ is approximately 2.0, with slight deviations introduced by curvature-induced modal dispersion at small bend radii. For the quasi-TM rings with $R = 15\,\mu$m ($n_g \approx 3.721$), the nominal expected FSR is ≈ 6.85 nm, subject to verification of the internal round-trip length of the `DoubleBus_Ring` cell. The loaded quality factor Q is given by:

$$Q = \frac{\lambda_0}{\Delta\lambda_{\mathrm{FWHM}}} = \frac{\pi n_g L_{\mathrm{rt}}}{\lambda_0} \frac{\sqrt{a t_1 t_2}}{1 - a t_1 t_2} \quad (21)$$

5 Waveguide Bragg Gratings

Waveguide Bragg gratings operate as distributed wavelength-selective optical reflectors [1]. In circuits `BraggGrating1` and `BraggGrating2`, periodic sidewall corrugations are implemented using the PCell `ebeam_bragg_te1310` on a 350 nm strip waveguide.

5.1 Bragg Reflection Condition

Light propagating along the corrugated waveguide experiences coherent back-scattering from each successive perturbation. Constructive interference occurs at the Bragg wavelength:

$$\lambda_{\mathrm{Bragg}} = 2\, n_{\mathrm{eff,mod}}(\lambda_{\mathrm{Bragg}})\, \Lambda \quad (22)$$

where $n_{\mathrm{eff,mod}}$ represents the modal index of the corrugation-modulated strip. In `BraggGrating1`, a grating period of $\Lambda = 317$ nm with corrugation width $\Delta w = 50$ nm and $N = 300$ periods is specified. Approximating $n_{\mathrm{eff,mod}}$ by the unperturbed strip index ($n_{\mathrm{eff}} \approx 2.43$ at

1310 nm, slightly modified by the corrugation), Eq. (22) places the fundamental stopband near $\lambda_{\text{Bragg}} \approx 1.54\,\mu$m, whereas centering a stopband strictly at 1310 nm would require $\Lambda \approx 270$ nm. The physical stopband position of the implemented cell is therefore directly evaluated from circuit-level ONA simulations in INTERCONNECT across a broad spectral sweep (1250–1650 nm) prior to measurement.

In `BraggGrating2`, the grating period is detuned to $\Lambda = 323$ nm with a reduced corrugation depth ($\Delta w = 35$ nm). The combined perturbations alter the Bragg resonance according to:

$$\Delta\lambda_{\text{Bragg}} \approx 2\, n_{\text{eff,mod}}\, \Delta\Lambda + 2\,\Lambda\, \Delta n_{\text{eff,mod}}(\Delta w) \tag{23}$$

where the period increment ($\Delta\Lambda = +6$ nm) induces an expected spectral red-shift of $\approx +29$ nm, while the narrower corrugation depth ($\Delta w = 35$ nm vs. 50 nm) reduces the coupling coefficient κ_{Bragg}, narrowing the reflection bandwidth.

5.2 Stopband Bandwidth and Coupling Coefficient

According to coupled-mode analysis for periodic media, the coupling coefficient per unit length κ_{Bragg} governs the reflectivity per period. The full-width photonic stopband bandwidth $\Delta\lambda_{\text{stopband}}$ is expressed as:

$$\Delta\lambda_{\text{stopband}} = \frac{\lambda_{\text{Bragg}}^2}{\pi n_g}\sqrt{\kappa_{\text{Bragg}}^2 + \left(\frac{\pi}{L_g}\right)^2} \tag{24}$$

where $L_g = N\Lambda$ is the physical length of the grating section ($N = 300$ periods). At the central Bragg resonance, the maximum power reflection approaches:

$$R_{\max} = \tanh^2(\kappa_{\text{Bragg}} L_g) \tag{25}$$

6 Cascaded Multi-Mode Waveguide Crossings

Complex photonic circuits demand low-loss, low-crosstalk waveguide intersections [1]. Standard orthogonal crossings suffer from diffraction losses and higher-order mode excitation at unguided intersections.

The sub-circuit `WaveguideCrossingCascade` incorporates a cascade of four compact crossings (`ebeam_crossing4`). Each crossing utilizes parabolic tapers expanding into a multimode interference (MMI) section that focuses the optical field into a waist at the center of the intersection, suppressing diffraction losses and crosstalk. By comparing the transmission of the four-crossing cascade against an adjacent baseline reference loopback waveguide, the insertion loss per intersection is estimated via:

$$\text{Loss}_{\text{crossing}} \approx \frac{\text{Loss}_{\text{cascade}}(\text{dB}) - \text{Loss}_{\text{ref}}(\text{dB})}{N_{\text{cross}}} \tag{26}$$

where $N_{\text{cross}} = 4$, assuming nominal reproducibility among identical grating couplers across the same exposure field. Since the insertion loss of an individual crossing is small ($\sim$ 0.1–0.2 dB), it is comparable to die-to-die grating coupler variability, so the extracted loss represents an estimate with associated experimental uncertainty.

7 Focusing Sub-Wavelength Grating Couplers (SWGCs)

Coupling optical energy between single-mode optical fibers (SMF-28, mode field diameter MFD $\approx 10.4\,\mu$m) and nanophotonic strip waveguides (500×220 nm) requires mode-area conversion across approximately 2.9 orders of magnitude (from $A_{\text{fiber}} \approx 85\,\mu\text{m}^2$ to $A_{\text{wg}} \approx 0.11\,\mu\text{m}^2$).

7.1 Bragg Diffraction and Off-Normal Coupling

The grating acts as an out-of-plane diffractive element. Phase matching in the longitudinal direction dictates that the guided wavevector β and the diffracted radiation wavevector satisfy the Bragg condition:

$$k_x = \beta - mK = \beta - m\frac{2\pi}{\Lambda} \tag{27}$$

where m is the diffraction order and $K = 2\pi/\Lambda$ represents the grating vector. For first-order diffraction ($m = 1$) into a cladding medium with index n_{clad} at an off-normal polar angle θ:

$$n_{\text{clad}} \sin\theta = n_{\text{eff,avg}} - \frac{\lambda_0}{\Lambda} \tag{28}$$

To suppress second-order back-reflections into the waveguide that degrade laser stability, couplers are engineered with an intentional tilt angle ($\theta \approx 8°$–$10°$).

7.2 Sub-Wavelength Engineering and Focusing Geometry

Standard fully-etched gratings suffer from high back-reflection at the silicon/oxide interface. The implemented sub-wavelength grating couplers (SWGCs) follow standard design principles established for the EBeam library [2]. For quasi-TE polarization (electric field oriented parallel to the grating lines), zeroth-order effective medium theory (EMT) predicts an intermediate synthetic effective index:

$$n_{\text{eff,EMT}} \approx \sqrt{f\, n_{\text{Si}}^2 + (1-f)\, n_{\text{SiO}_2}^2} \tag{29}$$

where f denotes the sub-wavelength duty cycle, mitigating modal impedance mismatch [2].

Furthermore, curved elliptical grating lines focus the diffracted wavefront directly into the 500 nm single-mode bus. The trajectory of the q-th grating line is defined by:

$$r_q(\phi) = \frac{q\,\lambda_0}{n_{\text{eff}} - n_{\text{clad}} \sin\theta\cos\phi} \tag{30}$$

where $q \in \mathbb{Z}^+$ designates the grating line index. The focusing geometry shortens the required taper length compared to a linear adiabatic taper.

8 Summary of Design Configurations

Table 2 summarizes the theoretical characteristics of all active cells embedded within the layout `EBeam_n-kostopoulos_v3.gds`.

Table 2: Comprehensive Photonic Cell Specifications

Sub-Circuit	Component Cell	Design Parameters	Target
MZI1	`ebeam_y_1550` + `ebeam_bdc_te1550`	$\Delta L \approx 140\,\mu$m, TE 1550	FSR 4.08 nm
MZI2	`ebeam_y_1550` + `ebeam_bdc_te1550`	$\Delta L \approx 170\,\mu$m, TE 1550	FSR 3.36 nm
Ring1	`ebeam_dc_halfring_straight`	$R = 10\,\mu$m, $g = 200$ nm, $L_c = 0$ (Add-Drop)	FSR 9.10 nm
Ring2	`ebeam_dc_halfring_straight`	$R = 5\,\mu$m, $g = 200$ nm, $L_c = 0$ (Add-Drop)	FSR 18.19 nm
SingleBus1	`DoubleBus_Ring`	$R = 15\,\mu$m, $g = 200$ nm (TM, terminated 2nd bus)	FSR 6.85 nm
SingleBus2	`DoubleBus_Ring`	$R = 15\,\mu$m, $g = 250$ nm (TM, terminated 2nd bus)	FSR 6.85 nm
Bragg1	`ebeam_bragg_te1310`	$\Lambda = 317$ nm, $\Delta w = 50$ nm, $N = 300$	λ_B from sim.
Bragg2	`ebeam_bragg_te1310`	$\Lambda = 323$ nm, $\Delta w = 35$ nm, $N = 300$	$\approx +29$ nm shift
Crossings	`ebeam_crossing4`	4× Cascaded MMI	Loss est.
Ref 1, 2, 3	Loopback Waveguide	TE 1550, TM 1550, TE 1310 (8-deg GC)	Baseline

References

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